

Algebraic Tori and Iwasawa Theory

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1 Introduction

Iwasawa theory arose from the study of the behavior of ideal class groups in cyclotomic towers of number fields. In the function field case, analogous questions are related to extensions of the field of constants of a curve over a finite field. In the classical case of abelian extensions of the field of rational numbers, the main results were obtained by Iwasawa, Washington, Mazur, and Wiles. Iwasawa developed the structure theory of modules over completed group algebras, established a connection with p -adic zeta functions, and formulated the Main Conjecture. Washington proved the vanishing of the μ -invariant in cyclotomic towers, while Mazur and Wiles proved the Main Conjecture in the abelian case. An exposition of these results can be found in the books of Washington [20] and Lang [7], as well as in the work of Mazur and Wiles [9]. Another part of this work is devoted to the arithmetic of algebraic tori. This theory studies invariants of tori over global fields, as well as invariants of their models over rings of integers of number fields or over curves. A substantial contribution to this area was made by Voskresenskii and his school. In particular, Voskresenskii's works studied Shafarevich–Tate groups of tori, weak approximation, and the exact sequences related to them (see [19]). On the other hand, Ono and his students studied formulas for class numbers of tori, including norm tori, and obtained important results on Tamagawa numbers of tori (see [17], [16], [13]).

The class group of a torus is a natural generalization of the ideal class group, since for the trivial torus $\mathbb{G}_m$ these notions coincide (see 4.1.2). It is therefore natural to attempt to transfer the ideas of Iwasawa theory from ideal class groups to class groups of algebraic tori. In other words, one is led to study the behavior of arithmetic invariants not only for $\mathbb{G}_m$, but also for more general tori in cyclotomic towers. The aim of this work is to initiate the study of this direction and to investigate the behavior of the Shafarevich–Tate group and the class group of algebraic tori in cyclotomic towers.

The following results were obtained in this work. In Subsection 3.6, we prove that the inverse limit of the Shafarevich–Tate groups of a torus in a cyclotomic tower is a finite module. It follows that, in the situation under consideration, the inverse limits of the class groups of the torus and of the first étale cohomology groups coincide (see 4.3.7). This makes it possible to describe elements of the corresponding limit as equivalence classes of torsors.

In the analysis of Voskresenskii's sequence, we obtain an example (see 3.5.7) showing that the sufficient conditions in one of Voskresenskii's propositions are not necessary. The same example also gives a counterexample to a statement formulated in [6].

The study of class groups of tori begins with the simplest function field case: a constant torus on a curve. In this case, we prove that the class group of the torus coincides with the first étale cohomology group (see 4.3.2). Then, in Subsection 6.3, for norm tori the class group is interpreted as the group of points of a twisted Jacobian of the curve. This interpretation allows one to observe that the growth of the class group of a constant norm torus on a curve is controlled by a certain L -function associated with the original curve and the torus (see 6.4).

2 Preliminaries

Here we recall some facts on algebraic tori. A more detailed exposition can be found in V. E. Voskresenskii's book *Algebraic Tori* (see [19]).

2.1 Algebraic Tori

For an arbitrary field K , denote by $\overline{K}$ and K_s its algebraic closure and separable closure, respectively. For an extension L/K , denote by $X_L = X \otimes_K L$ the L -scheme obtained from a K -scheme X by extension of scalars.

2.1.1 Definition. *A group scheme T over a field K is called a torus if*

$$T_{\overline{K}} \simeq \mathbb{G}_m^n,$$

where $n = \dim(T)$ is the dimension of T .

Let $\text{Gal}(K)$ be the absolute Galois group of the field K .

2.1.2 Definition. *The $\text{Gal}(K)$ -module of rational characters of a K -group scheme G is the $\text{Gal}(K)$ -module*

$$\text{Hom}_{K_s}(G_{K_s}, \mathbb{G}_m).$$

It is denoted by $\widehat{G}$ and defines a contravariant functor D from the category of group K -schemes to the category of continuous $\text{Gal}(K)$ -modules.

Tori over K can be described by the following theorem.

2.1.3 Theorem (see [19], Chapter III, Theorem 3.33). *When restricted to the category of K -group schemes of finite multiplicative type, the functor D defines an equivalence with the category of continuous $\text{Gal}(K)$ -modules of finite type.*

2.2 Weil restriction

Let L/K be a finite separable extension, and let X be an affine K -scheme of finite type. On the category of K -schemes, consider the functor to the category of sets defined on objects as follows:

$$\mathcal{X}(S) = X(S \otimes L).$$

It turns out that this functor is represented by an affine K -scheme, denoted by $R_{L/K}(X)$ (see [19, Chapter III, §5]).

2.2.1 Definition. *The Weil restriction of an L -scheme X is the K -scheme representing the functor $\mathcal{X}$.*

2.2.2 Remark. *If G is a group L -scheme, then $R_{L/K}(G)$ is a group K -scheme. Moreover, the groups $R_{L/K}(G)(K)$ and $G(L)$ are isomorphic.*

2.3 Norm tori

Let L/K be a finite separable extension of fields.

2.3.1 Construction. We construct the norm morphism

$$N: R_{L/K}(\mathbb{G}_m) \rightarrow \mathbb{G}_{m,K}.$$

To do this, it is enough to define a natural transformation of the functors of points of the corresponding schemes, since the functor

$$h: (K\text{-schemes}) \rightarrow \text{Fun}((K\text{-algebras}), (\text{sets})),$$

which sends a scheme to its functor of points, defines an equivalence between the category of K -schemes and a full subcategory of the category of functors (see [5, Proposition VI-2]). For a K -algebra A , using the description of the A -points of the Weil restriction (see 2.2.2), define N_A as follows:

$$N_A: (A \otimes L)^\times \rightarrow A^\times,$$

where the map is given by the formula

$$x \mapsto \det(m_x),$$

and m_x is the linear operator on $A \otimes L$ given by multiplication by the element $x \in (A \otimes L)^\times$.

2.3.2 Definition. The kernel of the norm morphism is called the norm torus and is denoted by $R_{L/K}^{(1)}(\mathbb{G}_m)$.

(For the definition of the kernel of a homomorphism of group schemes, see [19, Chapter I, beginning of §5].)

2.3.3 Remark. On K -points (see 2.2.2), the norm morphism coincides with the usual norm map in the extension L/K , and therefore

$$R_{L/K}^{(1)}(\mathbb{G}_m)(K) = \{x \in L^\times \mid N_{L/K}(x) = 1\}.$$

2.3.4 Proposition (see [19], Chapter VI, §8). Fix a basis $\{v_1, \dots, v_n\}$ of the extension L/K . Let

$$F(x_1, \dots, x_n) = N(x_1 v_1 + \dots + x_n v_n),$$

where $x_1 v_1 + \dots + x_n v_n$ is an element of $L[x_1, \dots, x_n]$, the free algebra on the set of generators $\{x_1, \dots, x_n\}$. Then

$$R_{L/K}^{(1)}(\mathbb{G}_m) \simeq \text{Spec}(K[x_1, \dots, x_n] / (F(x_1, \dots, x_n) = 1)).$$

The following lemma will be needed in Subsection 3.7.

2.3.5 Lemma. Consider the diagram of field extensions

$$\begin{array}{ccc} & L' & \\ L & \swarrow \quad \searrow & K' \\ & K & \end{array}.$$

Let L' be the compositum of the fields L and K' , and let K be their intersection. Then

$$R_{L'/K'}^{(1)}(\mathbb{G}_m) \simeq R_{L/K}^{(1)}(\mathbb{G}_m) \otimes_K K'.$$

Proof. Fix a basis $\{v_1, \dots, v_n\}$ of the extension L/K . Consider the polynomial F from Proposition 2.3.4 defining the norm torus corresponding to the extension L/K :

$$F(x_1, \dots, x_n) = N(x_1v_1 + \dots + x_nv_n).$$

Since L and K' are linearly disjoint, the elements $\{v_1, \dots, v_n\}$ can be taken as a basis of the extension L'/K' . Therefore the polynomial defining the norm torus corresponding to the extension L'/K' is equal to $N(x_1v_1 + \dots + x_nv_n)$ and hence is equal to the polynomial F . Thus, applying Proposition 2.3.4 twice, we obtain:

$$\begin{aligned} R_{L/K}^{(1)}(\mathbb{G}_m) \otimes_K K' &\simeq \operatorname{Spec}(K[x_1, \dots, x_n]/(F=1)) \otimes_K K' \simeq \\ &\simeq \operatorname{Spec}(K'[x_1, \dots, x_n]/(F=1)) \simeq R_{L'/K'}^{(1)}(\mathbb{G}_m). \end{aligned}$$

□

2.3.6 Proposition (see [19], Chapter IV, Example 4.10). *Let F be a Galois extension of the field K containing L , let $G = \operatorname{Gal}(F/K)$, and let $H = \operatorname{Gal}(F/L)$. Let $I = \operatorname{Ker}(\varepsilon)$, where*

$\varepsilon: \mathbb{Z}[G/H] \rightarrow \mathbb{Z}$ is the augmentation map given by

$$\sum_{i=1}^n a_i x_i \mapsto \sum_{i=1}^n a_i,$$

and $\{x_1, \dots, x_n\}$ is the basis of $\mathbb{Z}[G/H]$ consisting of cosets. Then

$$\widehat{R_{L/K}^{(1)}(\mathbb{G}_m)} = \operatorname{Hom}_{\mathbb{Z}}(I, \mathbb{Z}).$$

2.4 The weak approximation group

Let L/K be a Galois extension of global fields, and let T be a K -torus split by L .

2.4.1 Definition. *The weak approximation group of the torus T is defined as follows:*

$$A(T) = \left(\prod_v T(K_v) \right) / \overline{T(K)},$$

where the product is taken over all places of K , and $\overline{T(K)}$ is the topological closure of $T(K)$.

We say that T satisfies weak approximation if $A(T) = 0$.

2.4.2 Proposition (see [19], Chapter VI, Theorem 6.36). *If all decomposition groups in the extension L/K are cyclic, then $A(T) = 0$.*

The converse is false (see 3.5.7).

3 The Shafarevich–Tate group of a torus

3.1 Definition

Let L/K be a Galois extension of global fields with Galois group G , and let T be a K -torus split by L .

3.1.1 Definition. *The Shafarevich–Tate group of the torus T is defined as follows:*

$$\text{III}(T) = \text{Ker}\left(H^1(G, T(L)) \rightarrow H^1(G, T(A_L))\right),$$

where the map on cohomology is induced by the embedding $T(L) \hookrightarrow T(A_L)$, and A_L is the ring of adeles.

3.1.2 Proposition (see [19], Chapter VI, Corollary 6.27).

$$\text{III}(T) \simeq \text{Ker}\left(H^1(K, T(\overline{K})) \rightarrow \bigoplus_v H^1(K_v, T(\overline{K}_v))\right),$$

where the direct sum is taken over all places of K , and the map is the sum of the restriction homomorphisms.

In particular, $\text{III}(T)$ does not depend on the choice of a splitting field of the torus T .

3.1.3 Corollary (see [19], Proposition 6.25). *Let G_w be the decomposition group corresponding to a place w of the field L . Then*

$$\text{III}(T) \simeq \text{Ker}\left(H^1(G, T(L)) \rightarrow \bigoplus_v H^1(G_w, T(L_w))\right),$$

where in the direct sum, for each place v of the field K , one chooses a place w lying above v . For an archimedean place, the decomposition group is understood in the following sense: it is either $\mathbb{Z}/2\mathbb{Z}$ if the corresponding place is complex and lies above a real one, or 0 otherwise.

Examples of computations of Shafarevich–Tate groups of tori are given in Subsection 3.7.

3.2 Tate–Nakayama duality

In what follows, alongside ordinary group cohomology, we shall use Tate cohomology (see [3, Chapter VI]). For Tate cohomology we use the notation

$$\hat{H}^i(G, A),$$

where i is an integer, G is a group, and A is a G -module.

For $i \geq 1$, these groups coincide with the cohomology groups of G with the corresponding index, while for $i \leq -2$ they coincide with the homology groups of G with index $(-i - 1)$.

Denote by $A^\vee$ the group dual to A . In other words,

$$A^\vee = \text{Hom}(A, \mathbb{Q}/\mathbb{Z}).$$

3.2.1 Theorem (see [11], Chapter I, Corollary 2.3). *Let T be a torus over a local field K . Then there is an isomorphism*

$$\hat{H}^r(K, T(K_s)) \xrightarrow{\sim} \hat{H}^{2-r}(K, \hat{T})^\vee.$$

3.2.2 Definition. *Let T be a torus over a global field K . Define the idele class group of the torus T as follows:*

$$C_K(T) = T(A_K)/T(K).$$

3.2.3 Theorem (see [11], Chapter I, the discussion after Theorem 4.6). *Let T be a torus over a global field K split by a field L . Then there is an isomorphism*

$$\hat{H}^r(K, C_L(T)) \xrightarrow{\sim} \hat{H}^{2-r}(K, \hat{T})^\vee.$$

3.2.4 Remark. *If the K -tori in Theorems 3.2.1 and 3.2.3 are split by some finite Galois extension L , then the cohomology of the absolute Galois group in the statements can be replaced by the cohomology of the finite group $\text{Gal}(L/K)$.*

3.2.5 Corollary (see [11], Chapter I, Theorem 4.10). *Let L/K be a finite Galois extension, and let T be a torus over a global field K split by L . Then*

$$\text{III}(T) \simeq \text{Ker}[H^2(G, \hat{T}) \rightarrow \prod_v H^2(G_v, \hat{T})]^\vee.$$

Proof. In the long exact sequence of Tate cohomology associated with the short exact sequence

$$0 \rightarrow T(L) \rightarrow T(A_L) \rightarrow C_L(T) \rightarrow 0,$$

consider the fragment

$$\hat{H}^0(G, T(A_L)) \xrightarrow{\nu} \hat{H}^0(G, C_L(T)) \xrightarrow{\varphi} \hat{H}^1(G, T(L)) \xrightarrow{\psi} \hat{H}^1(G, T(A_L)).$$

Hence we obtain

$$\text{III}(T) \simeq \text{Ker}(\psi) \simeq \text{Coker}(\nu).$$

The cohomology of adelic points is computed in terms of local characteristics (see [19, Chapter VI, Proposition 6.25]):

$$\hat{H}^0(G, T(A_L)) \simeq \bigoplus_v \hat{H}^0(G_w, T(L_w)),$$

where in the direct sum, for each place v of the field K , one chooses a place w lying above v . By Tate–Nakayama duality (see 3.2.1 and 3.2.3),

$$\hat{H}^0(G_w, T(L_w)) \simeq \hat{H}^2(G_w, \hat{T})^\vee, \quad \hat{H}^0(G, C_L(T)) \simeq \hat{H}^2(G, \hat{T})^\vee.$$

Therefore

$$\text{III}(T) \simeq \text{Ker}[H^2(G, \hat{T}) \rightarrow \prod_v H^2(G_v, \hat{T})]^\vee.$$

□

3.2.6 Remark. *Since the character module is a finitely generated module over the group algebra of a finite group, it follows from the corollary that the corresponding cohomology groups are finite (see [4, Chapter IV, Theorem 7.1]). Therefore the Shafarevich–Tate group of a torus is finite.*

3.3 A method of computation for norm tori

The idea behind the method considered below for computing the Shafarevich–Tate group of norm tori corresponding to abelian extensions was found by the author in [6].

For the proof of Theorem 3.3.2, we shall need the following preliminary lemma.

3.3.1 Lemma. *For a finite abelian group G ,*

$$H^3(G, \mathbb{Z}) = \text{Hom}(\Lambda^2 G, \mathbb{Q}/\mathbb{Z}).$$

Proof. It is known (see [3, Chapter V, Theorem 6.4]) that

$$H_2(G, \mathbb{Z}) \simeq \Lambda^2 G.$$

Since $H_2(G, \mathbb{Z}) = \hat{H}^{-3}(G, \mathbb{Z})$, the product in Tate cohomology gives a duality (see [3, Chapter VI, §7])

$$\hat{H}^3(G, \mathbb{Z}) \simeq \hat{H}^{-3}(G, \mathbb{Z})^\vee,$$

whence

$$H^3(G, \mathbb{Z}) = \text{Hom}(\Lambda^2 G, \mathbb{Q}/\mathbb{Z}).$$

□

The following theorem gives a simple method for computing the Shafarevich–Tate group in the case of a norm torus.

3.3.2 Theorem. *Let $T = R_{L/K}^{(1)}(\mathbb{G}_m)$ be the norm torus corresponding to a finite abelian extension L/K of global fields with Galois group G . Then*

$$\text{III}(T) \simeq \text{Coker} \left[\prod_v (\Lambda^2 G_w) \rightarrow \Lambda^2 G \right],$$

where in the product, for each place v of the field K , one chooses a place w lying above v , and $\Lambda^2 G$ denotes the exterior square of the abelian group G .

Proof. For a norm torus, $\hat{T} = I^* = \text{Hom}_{\mathbb{Z}}(I, \mathbb{Z})$ (see 2.3.6). Consider the exact sequence

$$0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}[G] \rightarrow I^* \rightarrow 0,$$

dual to the exact sequence

$$0 \rightarrow I \rightarrow \mathbb{Z}[G] \rightarrow \mathbb{Z} \rightarrow 0.$$

Hence we observe that

$$H^i(G, \hat{T}) \simeq H^i(G, I^*) \simeq H^{i+1}(G, \mathbb{Z}).$$

Substituting this into the expression for $\text{III}(T)$ from Corollary 3.2.5, we obtain

$$\text{III}(T) \simeq \text{Ker} [H^3(G, \mathbb{Z}) \rightarrow \prod_v H^3(G_w, \mathbb{Z})]^\vee.$$

By Lemma 3.3.1, $H^3(G, \mathbb{Z}) = (\Lambda^2 G)^\vee$. Therefore

$$\text{III}(T) \simeq \text{Coker} \left[\prod_v \Lambda^2 G_w \rightarrow \Lambda^2 G \right].$$

□

3.4 Behavior under extension of scalars

Let K be a global field, let T be a K -torus, and let K' be a Galois extension of the field K . Denote by T' the torus obtained from T by extension of scalars to K' . In this situation, we construct restriction and corestriction maps on Shafarevich–Tate groups:

$$\text{res}: \text{III}(T) \rightarrow \text{III}(T'), \quad \text{cor}: \text{III}(T') \rightarrow \text{III}(T),$$

satisfying the relation $\text{cor} \circ \text{res} = [K' : K]$, and such that under successive extensions of scalars the resulting restriction and corestriction maps compose.

3.4.1 Construction. Consider the standard restriction and corestriction maps for the cohomology of the Galois groups of the fields K and K' (see [3, Chapter III, §3 and §9])

$$\text{res}: H^1(K, T(\overline{K})) \rightarrow H^1(K', T'(\overline{K'})), \quad \text{cor}: H^1(K', T'(\overline{K'})) \rightarrow H^1(K, T(\overline{K})),$$

as well as the corresponding maps for the cohomology of the Galois groups of all completions of these fields. Since these restriction and corestriction maps commute with the restriction maps to the cohomology of the Galois groups of completions (see [3, Chapter III, Proposition 9.5(iii)]), one can construct maps on the kernels of the restriction maps to the direct sums over all completions. We shall call these maps restriction and corestriction on the Shafarevich–Tate group. In other words, we have the following commutative diagram:

$$\begin{array}{ccccc} \text{III}(T) & \longrightarrow & H^1(K, T(\overline{K})) & \longrightarrow & \bigoplus_v H^1(K_v, T(\overline{K}_v)) \\ \uparrow \text{cores} & & \updownarrow & & \updownarrow \\ \text{III}(T') & \longrightarrow & H^1(K', T'(\overline{K'})) & \longrightarrow & \bigoplus_{v'} H^1(K'_{v'}, T'(\overline{K}'_{v'})) \end{array}$$

3.4.2 Proposition. The constructed restriction and corestriction maps on Shafarevich–Tate groups satisfy the relation

$$\text{cor} \circ \text{res} = [K' : K] \cdot \text{id},$$

and, under successive extensions of scalars corresponding to a tower of field extensions $K \subset K' \subset K''$, the composition of the restriction maps, respectively corestriction maps, corresponding to the extensions $K \subset K'$ and $K' \subset K''$ coincides with the restriction map, respectively corestriction map, for the extension $K \subset K''$.

Proof. The relation

$$\text{cor} \circ \text{res} = [K' : K] \cdot \text{id}$$

on cohomology (see [3, Chapter III, Proposition 9.5(ii)]) implies that the same relation holds for the constructed maps between Shafarevich–Tate groups. Also, since under successive extensions of scalars the restriction and corestriction maps on cohomology compose (see [3, Chapter III, Proposition 9.5(i)]), the same is true for the corresponding maps on Shafarevich–Tate groups. $\square$

3.4.3 Corollary. If the endomorphism of $\text{III}(T)$ given by multiplication by $[K' : K]$ is invertible, then the restriction map is injective and the corestriction map is surjective. In particular, this condition is satisfied when the torus T is split by a field L whose degree over K is coprime to $[K' : K]$.

Proof. The assumption implies that the map $\text{cor} \circ \text{res}$ is an automorphism; hence res is injective and cor is surjective. The final assertion follows from the fact that $\text{III}(T)$ is a finite group (see 3.2.6) annihilated by the order of the Galois group of the extension L/K , that is, by the degree $[L : K]$ (see [3, Chapter III, Proposition 10.1]). $\square$

3.4.4 Corollary. *Under extension of scalars of the torus T , as in Proposition 3.4.3, the order of the Shafarevich–Tate group does not decrease.*

Proof. By Remark 3.2.6, the Shafarevich–Tate group of the torus T is finite. Proposition 3.4.3 says that $\text{III}(T)$ embeds into the Shafarevich–Tate group of the torus obtained by extension of scalars. $\square$

3.5 Voskresenskii’s sequence

Let L/K be a Galois extension with Galois group G , and let T be a K -torus split by L .

3.5.1 Definition (see [19], Chapter IV, 4.34). *A G -module S is called a permutation G -module if it is isomorphic to a finite direct sum of modules of the form*

$$\mathbb{Z}[G/H],$$

where H is a subgroup of G , and G/H is the set of cosets.

G -modules M and N are called similar if

$$M \oplus S_1 \simeq N \oplus S_2,$$

where S_1, S_2 are permutation G -modules.

The similarity relation is an equivalence relation; the corresponding equivalence classes are called similarity classes.

3.5.2 Definition (see [19], Chapter IV, the discussion after Lemma 4.52). *For the character module of the torus T , consider an exact sequence of G -modules*

$$0 \rightarrow \hat{T} \rightarrow S \rightarrow N \rightarrow 0,$$

where S is permutation, and N is such that $\hat{H}^{-1}(H, N) = 0$ for every subgroup $H \subseteq G$. The similarity class of the module N is called the Picard class of the torus T and is denoted by $p(T)$. Such a sequence always exists, and the class $p(T)$ does not depend on the choice of the sequence (see [19, Chapter IV, Lemma 4.52 and the discussion after it]).

3.5.3 Remark. *Since the first Galois cohomology groups of similar modules are isomorphic, we define $H^1(L/K, p(T))$ as the cohomology of any representative.*

3.5.4 Theorem (see [19], Chapter VI, Theorem 6.38). *Let T be a torus over a global field K split by L . Then there exists an exact sequence of groups*

$$0 \rightarrow A(T) \rightarrow H^1(L/K, p(T))^\vee \rightarrow \text{III}(T) \rightarrow 0. \quad (1)$$

3.5.5 Theorem (see [19], Chapter VI, Theorem 6.46). *Let L/K be a Galois extension, and let T be the norm torus corresponding to this extension. Then*

$$H^1(L/K, p(T)) \simeq H^3(L/K, \mathbb{Z}).$$

In particular, Voskresenskii's exact sequence for a norm torus takes the following form:

$$0 \rightarrow A(T) \rightarrow H^3(L/K, \mathbb{Z})^\vee \rightarrow \text{III}(T) \rightarrow 0.$$

3.5.6 Corollary (see [19], Corollary 6.41). *If the torus T splits over a cyclic extension, then*

$$\text{III}(T) = 0 \text{ and } A(T) = 0.$$

Proof. For cyclic extensions, the group in the middle of the sequence 1 is equal to 0 (see [19, Chapter IV, Corollary 4.50]):

$$H^1(L/K, p(T)) = 0.$$

□

3.5.7 Example (Remark on 2.4.2). *Consider the number field L defined by the equation*

$$x^8 - x^7 - 4x^6 + 9x^5 + 11x^4 + 45x^3 - 100x^2 - 125x + 625.$$

It is a Galois extension of the field $\mathbb{Q}$ with Galois group $\mathbb{Z}/4 \times \mathbb{Z}/2$, and it coincides with the compositum of its subfields $\mathbb{Q}(\zeta_5)$ and $\mathbb{Q}(\sqrt{-19})$.

In this extension, only 5, 19, and the archimedean place ramify, with decomposition groups $\mathbb{Z}/4$, $\mathbb{Z}/2 \times \mathbb{Z}/2$, and $\mathbb{Z}/2\mathbb{Z}$, respectively. The map

$$\Lambda^2(\mathbb{Z}/2 \times \mathbb{Z}/2) \rightarrow \Lambda^2(\mathbb{Z}/4 \times \mathbb{Z}/2)$$

is zero. The maps corresponding to the remaining primes are also zero, since in these cases the decomposition groups are cyclic and their exterior square is zero. Therefore, by Theorem 3.3.2,

$$\text{III}(T) = \mathbb{Z}/2.$$

By Lemma 3.3.1, we obtain

$$H^3(\mathbb{Z}/4 \times \mathbb{Z}/2, \mathbb{Z}) = \Lambda^2(\mathbb{Z}/4 \times \mathbb{Z}/2) = \mathbb{Z}/2.$$

Hence Voskresenskii's sequence implies that $A(R_{L/\mathbb{Q}}^{(1)}) = 0$. At the same time, the extension has a decomposition group corresponding to 19 which is not cyclic.

This example is a counterexample to the statement formulated in [6, Lemma 6.7].

3.6 Application to tori in cyclotomic towers

Let T be a $\mathbb{Q}$ -torus split by a field L . Consider the ℓ -cyclotomic tower, where ℓ is an odd prime:

$$\mathbb{Q} \subset \mathbb{Q}(\zeta_\ell) \subset \mathbb{Q}(\zeta_{\ell^2}) \subset \cdots \subset \mathbb{Q}(\zeta_{\ell^\infty}).$$

Denote

$$K_n = \mathbb{Q}(\zeta_{\ell^n}), \quad L_n = L(\zeta_{\ell^n}),$$

and also

$$K_\infty = \bigcup_{n \geq 0} K_n, \quad L_\infty = \bigcup_{n \geq 0} L_n.$$

Assume that L is such that L and K_n are linearly disjoint for all n , and let T_n be the torus obtained from T by extension of scalars to K_n . It follows from the assumption that T_n is split by the field L_n . In this case, for $n \geq m$ the following maps arise (see 3.4):

$$\text{III}(T_m) \xrightarrow{\text{res}_m^n} \text{III}(T_n), \quad \text{III}(T_n) \xrightarrow{\text{cor}_m^n} \text{III}(T_m).$$

3.6.1 Remark. *If the degrees $[L : K]$ and $[K_n : K]$ are coprime, then Proposition 3.4.3 implies that in the tower the order of the group $\text{III}(T_n)$ does not decrease as n grows.*

Voskresenskii's sequence (see 1) implies that the groups $\text{III}(T_n)$ are bounded.

3.6.2 Corollary. *The orders of the groups $\text{III}(T_n)$ are bounded by the order of the group $H^1(L/K, p(T))$, which does not depend on n .*

The corestriction maps in this situation form an inverse system, since they compose under successive extensions of scalars (see 3.4.2); that is, $\text{cor}_l^m \circ \text{cor}_m^n = \text{cor}_l^n$ for $n \geq m \geq l$. Denote by $\text{Tate}_\ell(\text{III}(T))$ the limit of the inverse system obtained from the system described above by restricting to the ℓ -primary parts of the groups $\text{III}(T_i)$. The preceding corollary then says that $\text{Tate}_\ell(\text{III}(T))$ is a finite $\mathbb{Z}_\ell[[\text{Gal}(K_\infty/\mathbb{Q})]]$ -module, where $\mathbb{Z}_\ell[[\text{Gal}(K_\infty/\mathbb{Q})]]$ is the completed group algebra of the group $\text{Gal}(K_\infty/\mathbb{Q})$ over the ring $\mathbb{Z}_\ell$.

3.7 Computations of Shafarevich–Tate groups of tori

Let T be the norm torus corresponding to the extension L/K , and let T_n be the torus defined in Subsection 3.6. By Lemma 2.3.5, the torus T_n obtained by base change is isomorphic to the norm torus corresponding to the extension L_n/K_n . In other words,

$$T_n \simeq R_{L_n/K_n}^{(1)}(\mathbb{G}_m).$$

Unlike the group $H^1(L/K, p(T))$, which is isomorphic to $H^3(L/K, \mathbb{Z})$, the behavior of the groups $\text{III}(T)$ and $A(T)$ depends not only on the Galois group, but also on the arithmetic of the extension L/K .

3.7.1 Example. *Consider the following situation:*

$$L = \mathbb{Q}(\sqrt{13}, \sqrt{17}), \quad K = \mathbb{Q},$$

where ℓ is a prime as in Subsection 3.6. In this case, all decomposition groups are cyclic. From Proposition 2.4.2, we obtain $A(T) = 0$. To compute the group in the middle of Voskresenskii's sequence (see 1), we use Lemma 3.3.1, which gives

$$H^1(L/K, p(T)) \simeq H^3(L/K, \mathbb{Z}) \simeq \Lambda^2(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}) \simeq \mathbb{Z}/2\mathbb{Z}.$$

It then follows from Voskresenskii's sequence that $\text{III}(T) \simeq \mathbb{Z}/2\mathbb{Z}$. The same argument shows that $\text{III}(T_1) \simeq \mathbb{Z}/2\mathbb{Z}$. Since the Shafarevich–Tate group of the torus on the first level of the tower embeds into all groups $\text{III}(T_n)$ on higher levels, and their orders are bounded by the order of $H^1(L/K, p(T))$, we obtain that for every n

$$\text{III}(T) \simeq \text{III}(T_n) \simeq \mathbb{Z}/2\mathbb{Z}.$$

3.7.2 Example. Consider

$$L = \mathbb{Q}(\sqrt{-7}, \sqrt{13}), \quad K = \mathbb{Q},$$

and let $\ell = 3$. In this case, the decomposition groups of the primes lying above 7 and 13 are equal to

$$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

while all other decomposition groups are cyclic. Therefore, by Theorem 3.3.2, we obtain $\text{III}(T) \simeq 0$. However, since $7 \equiv 1 \pmod{3}$ and $13 \equiv 1 \pmod{3}$, the primes 7 and 13 split completely in the extension K_1 , so on the first level all decomposition groups become cyclic. Then, by Theorem 3.3.2, $\text{III}(T_1) \simeq \mathbb{Z}/2\mathbb{Z}$. Thus, since the Shafarevich–Tate groups on higher levels contain the groups from lower levels, we obtain

$$\text{III}(T) \simeq 0, \text{III}(T_1) \simeq \text{III}(T_n) \simeq \mathbb{Z}/2\mathbb{Z}.$$

4 The class group of a torus

4.1 Definition

4.1.1 Definition. Let K be a number field. The class group of a K -torus T is defined as the quotient

$$\text{Cl}(T) = T(A_K)/T(K)T(A_K^\infty),$$

where A_K is the ring of adeles, and A_K^∞ is the subring of adeles whose non-archimedean components lie in the rings of integers of the corresponding completions.

If K is a function field, then the class group of the torus T is defined as the quotient

$$\text{Cl}(T) = T^{(1)}(A_K)/T(K),$$

where

$$T^{(1)}(A_K) = \{a \in T(A_K) \mid \chi(a) \in I_K^{(1)} \text{ for all } \chi \in \widehat{T} \text{ defined over } K\},$$

and $I_K^{(1)}$ is the group of ideles of norm one.

These quotients are finite (see [19, Chapter VI, the discussion after Theorem 6.29]). Denote their orders by h_T .

4.1.2 Example. From the adelic description of the ideal class group of a number field K , it follows that

$$\text{Cl}(\mathbb{G}_{m,K}) = I_K/K^\times I_K^\infty = \text{Cl}(K),$$

where I_K is the idele group of K .

Similarly, in the case of a function field K , from the adelic description of the group of divisor classes of degree zero, we obtain

$$\text{Cl}(\mathbb{G}_{m,K}) = I_K^{(1)}/K^\times = \text{Cl}^0(C),$$

where C is the smooth model of the curve corresponding to the field K , and $\text{Cl}^0(C)$ is the group of divisor classes of degree zero on this curve.

4.1.3 Proposition ([19], Proposition 6.31). Let L/K be an extension of global fields, and let T be an L -torus. Then for the torus obtained from T by Weil restriction, the following assertion holds:

$$\text{Cl}(R_{L/K}(T)) = \text{Cl}(T).$$

4.2 Behavior under extension of scalars

To construct the norm map on class groups of tori, we need to generalize the construction of the norm morphism, previously defined for $\mathbb{G}_m$ (see 2.3.1), to the case of an arbitrary torus.

4.2.1 Construction. *Let K'/K be a finite separable extension of fields, let T be a K -torus, and let T' be the extension of scalars of T to K' . In this situation, we construct the norm morphism*

$$N: R_{K'/K}(T') \rightarrow T.$$

To do this, we shall regard the corresponding algebraic groups as sheaves of abelian groups on the étale site $\mathrm{Spec}(K)_{\mathrm{\acute{e}t}}$. Denote by f the morphism corresponding to the field extension. It is étale, and therefore there is a pair of adjoint functors between the categories of sheaves of abelian groups (see [1, Chapter IX, §5])

$$f_! : \mathrm{Ab}((\mathrm{Spec} K')_{\mathrm{\acute{e}t}}) \rightleftarrows \mathrm{Ab}((\mathrm{Spec} K)_{\mathrm{\acute{e}t}}) : f^*, \quad f_! \dashv f^*,$$

where f^ is the inverse image functor, and $f_!$ is the extension by zero functor. Since the morphism f is finite étale, we have $f_! \simeq f_*$, where f_* is the direct image functor. Therefore*

$$f_!(T') \simeq f_*(T') \simeq R_{K'/K}(T').$$

The counit of the adjunction $f_! \dashv f^$ gives a homomorphism of sheaves on the small étale site*

$$\varepsilon: f_! f^* T \simeq f_* f^* T = R_{K'/K}(T') \rightarrow T.$$

Hence, for every separable extension F/K , we obtain the desired homomorphism on F -points. Since a morphism of smooth algebraic groups is determined by its action on F -points, we obtain the desired morphism of algebraic groups.

4.2.2 Remark. *Using the unit of the adjunction $f^* \dashv f_*$, one can construct a map in the opposite direction:*

$$\mathrm{res}: T \rightarrow R_{K'/K}(T'),$$

called the restriction map. The composition $N \circ \mathrm{res}$ is equal to multiplication by the degree of the extension $[K' : K]$ (see [1, Chapter IX, §5]).

For an arbitrary global field K , denote by $V(K)$ the set of equivalence classes of valuations of K .

4.2.3 Definition. *Let K be a global field, let T be a K -torus, and let K'/K be a finite separable extension of fields. Denote by T' the torus obtained from T by extension of scalars to K' . The norm map on class groups of tori,*

$$N: \mathrm{Cl}(T') \rightarrow \mathrm{Cl}(T),$$

is defined on representatives of classes of adelic points as follows:

$$\left[(a_w)_{w \in V(K')} \right] \mapsto \left[(a_v)_{v \in V(K)} \right],$$

where the components of the adelic point are defined by

$$a_v = \prod_{w|v} (N_{w/v} a_w),$$

and $N_{w/v}$ is the norm morphism for the torus T_{K_v} corresponding to the extension K'_w/K_v , considered on K_v -points, that is,

$$N_{w/v}: T'(K'_w) \rightarrow T(K_v).$$

This definition is correct, since the norm sends $T'(K')$ to $T(K)$; in the number field case, it sends $T'(A_K^\infty)$ to $T(A_K^\infty)$, and in the function field case, it sends $T'^{(1)}(A_{K'})$ to $T^{(1)}(A_K)$.

4.2.4 Remark. Similarly, using the restriction map

$$\text{res}: T \rightarrow R_{K'/K}(T'),$$

one can construct a restriction map between class groups of tori:

$$\text{res}: \text{Cl}(T) \rightarrow \text{Cl}(T').$$

The composition $N \circ \text{res}$ is equal to the map of multiplication by the degree of the extension $[K' : K]$, by the construction of N and res .

The norm maps compose under successive extensions of scalars. Consider the situation as in Subsection 3.6. The norm maps on class groups of tori form an inverse system; we denote the limit of the ℓ -primary parts of these groups by

$$\text{Tate}_\ell(\text{Cl}(T)).$$

4.3 Relation with Nisnevich and étale cohomology

Let S be the spectrum of a Dedekind ring with field of fractions K . We will be interested in the cases where S is the spectrum of the ring of integers of a number field or an open affine subscheme of a curve over a finite field. Let $\mathcal{T}$ be a group scheme of finite type over S such that its generic fiber is the torus T . We shall call such a scheme an integral model of the torus T .

4.3.1 Definition. The class group of an integral model of a torus $\mathcal{T}$ is defined as follows:

$$\text{Cl}(\mathcal{T}) = \mathcal{T}(A_K)/\mathcal{T}(K)\mathcal{T}(A_K^S),$$

where A_K is the ring of adeles, and A_K^S is the subring of S -integral adeles.

We denote by $H^n(X_{\text{cd}}, F)$ the Nisnevich cohomology of a scheme X with coefficients in a sheaf F on the site X_{cd} (see [15]), and by $H^n(X_{\text{ét}}, G)$ the étale cohomology of X with coefficients in a sheaf G on the site $X_{\text{ét}}$ (see [10]).

4.3.2 Theorem (see [14], Theorem 1). Let $\mathcal{T}$ be an integral model of a torus T . Then there is the following isomorphism:

$$\text{Cl}(\mathcal{T}) \simeq H^1(S_{\text{cd}}, \mathcal{T}).$$

Now take as a model of the K -torus T over S the connected component of the identity in its lft Néron model ([2, Chapter 10, §5, Proposition 6]). We denote this group scheme by $\mathcal{T}$. We construct the norm morphism for it.

4.3.3 Construction. *Let K'/K be a finite separable extension of global fields, and let S' be the integral closure of S in K' . We construct in this situation the norm morphism*

$$N: \mathcal{R}_{S'/S}(\mathcal{T}') \rightarrow \mathcal{T},$$

where $\mathcal{T}'$ is the connected component of the identity in the lft Néron model of the K' -torus $T' = T_{K'}$, and $\mathcal{R}_{S'/S}(\mathcal{T}')$ is the scheme obtained by Weil restriction of $\mathcal{T}'$, which coincides with the connected component of the identity in the lft Néron model of the torus $R_{K'/K}(T')$ (see [2, Chapter 7, §6, Proposition 6]). By the Néron mapping property for the lft Néron model (see [2, Chapter 1, §2, Definition 1]), applied to the norm morphism for tori over fields, we obtain

$$N: \mathcal{R}_{S'/S}(\mathcal{T}')_{\text{lft}} \rightarrow \mathcal{T}_{\text{lft}},$$

where $\mathcal{R}_{S'/S}(\mathcal{T}')_{\text{lft}}$ and $\mathcal{T}_{\text{lft}}$ are the lft Néron models of the tori $R_{K'/K}(T')$ and T , respectively. Since the original norm morphism is a homomorphism of algebraic groups over K , the resulting norm morphism on integral models is also a homomorphism of group schemes, and therefore this homomorphism descends to a morphism on the connected components of the identities.

Let K be a global field, let T be a K -torus, let K'/K be a finite separable extension of fields, and let $T' = T_{K'}$. Let $\mathcal{T}$ and $\mathcal{T}'$ be the connected components of the identities in the lft Néron models of the tori T and T' .

4.3.4 Definition. *The norm map on étale cohomology,*

$$N: H^n(S'_{\text{ét}}, \mathcal{T}') \rightarrow H^n(S_{\text{ét}}, \mathcal{T}),$$

is defined as the composition of the following maps:

$$H^n(S'_{\text{ét}}, \mathcal{T}') \simeq H^n(S_{\text{ét}}, \mathcal{R}_{S'/S}(\mathcal{T}')) \xrightarrow{H_{\text{ét}}^n(N)} H^n(S_{\text{ét}}, \mathcal{T}).$$

Similarly, using Theorem 6.1.1, we define the norm on class groups of models of the torus,

$$N: \text{Cl}(\mathcal{T}') \rightarrow \text{Cl}(\mathcal{T}),$$

as follows:

$$\text{Cl}(\mathcal{T}') \simeq H^1(S'_{\text{cd}}, \mathcal{T}') \simeq H^1(S_{\text{cd}}, \mathcal{R}_{S'/S}(\mathcal{T}')) \xrightarrow{H_{\text{cd}}^1(N)} H^1(S_{\text{cd}}, \mathcal{T}) \simeq \text{Cl}(\mathcal{T}).$$

As in the previous cases where the norm map appeared, norms compose under successive base change. This follows from the uniqueness requirement for the extension of a morphism in the Néron mapping property. Consider the situation as in Subsection 3.6. The norm maps on first étale cohomology groups and on class groups form inverse systems. We denote the limits of the ℓ -primary parts of these groups by

$$\text{Tate}_{\ell}(H^1(S_{\text{ét}}, \mathcal{T})) \text{ and } \text{Tate}_{\ell}(\text{Cl}(\mathcal{T})).$$

The class group of the integral model $\mathcal{T}$, the first étale cohomology of the base with coefficients in $\mathcal{T}$, and the Shafarevich–Tate group of the generic fiber T are related by the following theorem.

4.3.5 Theorem ([14], Theorem 2). *There is a short exact sequence:*

$$0 \rightarrow \text{Cl}(\mathcal{T}) \rightarrow H^1(S_{\text{ét}}, \mathcal{T}) \rightarrow \text{III}(T) \rightarrow 0.$$

4.3.6 Definition. *Modules M and N over a ring will be called pseudo-isomorphic if there exists a homomorphism $M \rightarrow N$ whose kernel and cokernel are finite.*

Let χ be a character of $\text{Gal}(K_1/\mathbb{Q})$. Denote the χ -isotypic components of the limits of étale cohomology groups and class groups as follows:

$$\text{Tate}_\ell(H^1(S_{\text{ét}}, \mathcal{T}))_\chi \text{ and } \text{Tate}_\ell(\text{Cl}(\mathcal{T}))_\chi.$$

The map $\text{Cl}(\mathcal{T}) \rightarrow H^1(S_{\text{ét}}, \mathcal{T})$ is obtained from the comparison functor between the Nisnevich topology and the étale topology on S (see [15, sequence 1.36.3]). Since the constructed norm maps on the class group and on the first étale cohomology groups arise from the same norm morphism of group schemes, they commute with the comparison functor and, consequently, with the map $\text{Cl}(\mathcal{T}) \rightarrow H^1(S_{\text{ét}}, \mathcal{T})$. The norm maps on first étale cohomology groups and on the Shafarevich–Tate group commute with the map $H^1(S_{\text{ét}}, \mathcal{T}) \rightarrow \text{III}(T)$, since $\text{III}(T)$ can be viewed as a subgroup of the étale cohomology of the generic point of S with coefficients in the generic fiber. Therefore, for the limits we have an exact sequence

$$0 \rightarrow \text{Tate}_\ell(\text{Cl}(\mathcal{T})) \rightarrow \text{Tate}_\ell(H^1(S_{\text{ét}}, \mathcal{T})) \rightarrow \text{Tate}_\ell(\text{III}(T)),$$

which is not necessarily exact on the right. Nevertheless, since $\text{Tate}_\ell(\text{III}(T))$ is a finite module over $\mathbb{Z}_\ell[[\text{Gal}(K_\infty/\mathbb{Q})]]$, the cokernel of the following map:

$$\text{Tate}_\ell(\text{Cl}(\mathcal{T})) \rightarrow \text{Tate}_\ell(H^1(S_{\text{ét}}, \mathcal{T}))$$

is also finite. Thus, the following corollary follows from Theorem 4.3.5.

4.3.7 Corollary. *Let χ be a character of $\text{Gal}(K_1/\mathbb{Q})$. Then the $\mathbb{Z}_\ell[[\text{Gal}(K_\infty/K_1)]]$ -modules*

$$\text{Tate}_\ell(\text{Cl}(\mathcal{T}))_\chi \text{ and } \text{Tate}_\ell(H^1(S_{\text{ét}}, \mathcal{T}))_\chi$$

are pseudo-isomorphic.

5 The class group of a torus in the number field case

5.1 The case of a norm torus

Consider norm tori $T = R_{L/K}^{(1)}(\mathbb{G}_m)$ corresponding to a Galois extension of global fields L/K with Galois group G . In this case, the order of the class group of the torus is computed by a formula due to Ono (see [17]).

5.1.1 Theorem ([17], Theorem 2.1).

$$\frac{h_L}{h_T h_K} = \frac{|\text{III}(T)| \cdot \prod_v |H^0(G_w, \mathcal{O}_w^\times)|}{[L' : K] \cdot |H^0(G, \mathcal{O}_L)|},$$

where L' is the maximal abelian subextension of L/K , and in the product, for each v , one chooses a place w above v .

5.2 Norm tori of imaginary quadratic extensions

Consider the simplest norm tori, namely those corresponding to imaginary quadratic extensions $\mathbb{Q}(\sqrt{-m})/\mathbb{Q}$. Let ℓ be a prime for which we consider the $\mathbb{Z}_\ell$ -extension of $\mathbb{Q}(\sqrt{-m}, \zeta_\ell)$.

In this case, the formulas simplify to the following form:

$$h(n) = h(R_{L_n/K_n}^{(1)}(\mathbb{G}_m)) = \frac{h(L_n)}{h(K_n)2^{\tau(L_n/K_n)-\varepsilon(L_n/K_n)-1}},$$

where $\tau(n) = \tau(L_n/K_n)$ is the number of places which ramify, and

$$\varepsilon(n) = \varepsilon(L_n/K_n) = v_2\left(\hat{H}^0(G_n, \mathcal{O}_{L_n}^\times)\right).$$

5.3 Computations

The following table was obtained using the computer algebra package PARI/GP [18] and contains experimental data on jumps at the first levels of the cyclotomic tower for the class group of the norm torus corresponding to an imaginary quadratic field. The notation is taken from Subsection 5.2.

Table 1: Values of the invariants of the norm torus at the first levels of the ℓ -cyclotomic tower.

m	n	$\varepsilon(n)$	$\tau(n)$	$h(L_n)$	$h(K_n)$	$h(T_n)$
1	1	0	1	1	1	1
1	2	0	1	1	1	1
2	1	0	1	1	1	1
2	2	0	1	3	1	3
5	1	1	2	2	1	2
5	2	1	2	6	1	6
7	1	1	2	1	1	1
7	2	1	2	1	1	1
10	1	1	2	2	1	2
10	2	1	2	14	1	14
11	1	0	1	1	1	1
11	2	0	1	3	1	3
13	1	1	3	4	1	2
13	2	1	3	52	1	26
14	1	1	3	4	1	2
14	2	1	3	36	1	18
17	1	1	2	4	1	4
17	2	3	4	48	1	48
19	1	1	2	1	1	1
19	2	3	6	4	1	1
22	1	1	2	2	1	2
22	2	1	2	38	1	38

m	n	$\varepsilon(n)$	$\tau(n)$	$h(L_n)$	$h(K_n)$	$h(T_n)$
23	1	0	1	3	1	3
23	2	0	1	9	1	9
26	1	0	3	12	1	3
26	2	0	3	252	1	63
29	1	1	2	6	1	6
29	2	1	2	126	1	126
31	1	1	2	3	1	3
31	2	1	2	9	1	9
34	1	1	2	4	1	4
34	2	3	4	112	1	112
35	1	1	3	2	1	1
35	2	1	3	18	1	9
37	1	1	3	4	1	2
37	2	3	7	208	1	26
38	1	1	3	6	1	3
38	2	3	7	288	1	36
41	1	1	2	8	1	8
41	2	1	2	216	1	216
43	1	1	2	1	1	1
43	2	1	2	13	1	13

6 The class group of a torus in the geometric case

Let K be the function field of a smooth projective curve C defined over a finite field $k = \mathbb{F}_q$, where q is a power of a prime number p .

6.1 Relation with étale cohomology and Nisnevich cohomology

Let now $\mathcal{T}$ be a constant torus on the curve C , that is, the torus obtained by base change of a k -torus T to C . A group scheme on the curve C naturally defines sheaves, represented by the scheme $\mathcal{T}$, on the sites $C_{\text{ét}}$ and C_{cd} .

6.1.1 Theorem. *Let C be a smooth projective curve over a finite field k , let x be a k -point of C , and let $\mathcal{T}$ be a constant torus on the curve C obtained from a k -torus T . Then*

$$\text{Cl}(\mathcal{T}) \times \mathbb{Z}^r \simeq H^1(C_{\text{cd}}, \mathcal{T}),$$

where r is the dimension of the maximal split subtorus of T , and $\text{Cl}(\mathcal{T})$ is the class group of the restriction of $\mathcal{T}$ to the open affine subscheme $S = C \setminus \{x\}$.

Proof. Consider the following fragment of the long exact cohomology sequence for the pair $(C_{\text{cd}}, S_{\text{cd}})$ (see [15, sequence 1.23.3]):

$$H^0(C_{\text{cd}}, \mathcal{T}) \rightarrow H^0(S_{\text{cd}}, \mathcal{T}) \rightarrow H_x^1(C_{\text{cd}}, \mathcal{T}) \rightarrow H^1(C_{\text{cd}}, \mathcal{T}) \rightarrow H^1(S_{\text{cd}}, \mathcal{T}) \rightarrow 0.$$

The map $H^0(C_{\text{cd}}, \mathcal{T}) \rightarrow H^0(S_{\text{cd}}, \mathcal{T})$ is surjective. We show that $H_x^1(C_{\text{cd}}, \mathcal{T}) \simeq \mathbb{Z}^r$. It is known (see [15, Corollary 1.28]) that

$$H_x^1(C_{\text{cd}}, \mathcal{T}) \simeq H_x^1((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T}).$$

Consider the following fragment of the long exact cohomology sequence for the pair $((\mathcal{O}_{C,x}^h)_{\text{cd}}, \{\eta\}_{\text{cd}})$, where η is the generic point of $\text{Spec}(\mathcal{O}_{C,x}^h)$:

$$H^0((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T}_x) \rightarrow H^0((k(\eta))_{\text{cd}}, \mathcal{T}_x) \rightarrow H_x^1((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T}_x) \rightarrow H^1((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T}_x).$$

The cohomology group $H^1((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T})$ is zero, since $\mathcal{O}_{C,x}^h$ is a Henselian local ring (see [15, Lemma 1.18.1]). Therefore we obtain

$$H_x^1((\mathcal{O}_{C,x}^h)_{\text{cd}}, \mathcal{T}_x) \simeq \frac{\mathcal{T}(k(\eta))}{\mathcal{T}(\mathcal{O}_{C,x}^h)},$$

where the right-hand side is expressed as follows (see [19, Corollary 6.11]):

$$\frac{\mathcal{T}(k(\eta))}{\mathcal{T}(\mathcal{O}_{C,x}^h)} \simeq \mathbb{Z}^r.$$

Thus, since by Theorem 4.3.2 we have $H^1(S_{\text{cd}}, \mathcal{T}) \simeq \text{Cl}(\mathcal{T})$, we obtain the following short exact sequence:

$$0 \rightarrow \mathbb{Z}^r \rightarrow H^1(C_{\text{cd}}, \mathcal{T}) \rightarrow \text{Cl}(\mathcal{T}) \rightarrow 0,$$

which splits by the degree map. $\square$

The Nisnevich cohomology of the curve C with coefficients in the sheaf $\mathcal{T}$ coincides with its étale cohomology.

6.1.2 Theorem.

$$H^1(C_{\text{cd}}, \mathcal{T}) \simeq H^1(C_{\text{ét}}, \mathcal{T})$$

Proof. It is enough to check that $H^1(\mathcal{O}_{C,x}^h, \mathcal{T}_x^h) = 0$ for every scheme-theoretic point x of C (see [15, Lemma 1.38]), where $\mathcal{O}_{C,x}^h$ is the Henselization of the local ring $\mathcal{O}_{C,x}$, and $\mathcal{T}_x^h$ is the stalk of the sheaf corresponding to $\mathcal{T}$. We use the fact that

$$H^1\left((\mathcal{O}_{C,x}^h)_{\text{ét}}, \mathcal{T}_x^h\right) = (\mathcal{T}_x^h\text{-torsors over } \mathcal{O}_{C,x}^h).$$

If x is not the generic point, then by Lang's theorem (see [12, Corollary 17.98]), algebraic groups over finite fields have no nontrivial torsors. In other words, every torsor has a section. The group scheme $\mathcal{T}_x^h$ and its reduction at the point x are smooth algebraic groups, since the torus $\mathcal{T}$ is constant. By Hensel's lemma for Henselian rings (see [2, Chapter 2, Section 3, Proposition 5]), a section of a torsor over the residue field can be lifted to a section over $\mathcal{O}_{C,x}^h$. Hence $\mathcal{T}_x^h$ has no nontrivial torsors over $\mathcal{O}_{C,x}^h$, and

$$H^1\left((\mathcal{O}_{C,x}^h)_{\text{ét}}, \mathcal{T}_x^h\right) = 0.$$

Over the generic point η , since the torus is constant, it becomes split after extending the base finite field of the curve. Let $k(\eta)$ be the residue field at the generic point, and let $k'(\eta)$ be the residue field at the generic point on the curve after extending the base field to a finite field k' over which the torus $\mathcal{T}$ splits. In this case, there is an exact restriction-inflation sequence for Galois cohomology (see [4, Chapter IV, Proposition 5.1])

$$0 \rightarrow H^1(k'/k, \mathcal{T}) \rightarrow H^1(k(\eta), \mathcal{T}) \rightarrow H^1(k'(\eta), \mathcal{T}).$$

By Lang's theorem, $H^1(k'/k, \mathcal{T}) = 0$. Since over $k'(\eta)$ the torus $\mathcal{T}$ is split, we have $H^1(k'(\eta), \mathcal{T}) \simeq H^1(k'(\eta), \mathbb{G}_m^n) = 0$, where the last equality is Hilbert's Theorem 90. From the exact sequence above, we obtain $H^1(k(\eta), \mathcal{T}) = 0$. $\square$

6.2 Constant norm tori

Let now $\mathcal{T}$ be a constant norm torus on the curve C corresponding to an extension of the base field l/k . We shall need the fact that the norm morphism is a surjection of sheaves in the étale topology.

6.2.1 Theorem (see [13]). *The norm morphism*

$$N : R_{C_l/C}(\mathbb{G}_m) \rightarrow \mathbb{G}_m$$

is a surjection of étale sheaves represented by the corresponding schemes.

6.2.2 Corollary.

$$\mathrm{Cl}(\mathcal{T}) \simeq \mathrm{Ker} [N : \mathrm{Pic}(C_l) \rightarrow \mathrm{Pic}(C)]$$

Proof. Consider the fragment of the long exact cohomology sequence associated with the following short exact sequence

$$0 \rightarrow R_{L/K}^{(1)}(\mathbb{G}_m) \rightarrow R(\mathbb{G}_m) \rightarrow \mathbb{G}_m \rightarrow 0.$$

It has the form

$$H^0(R(\mathbb{G}_m)) \rightarrow H^0(\mathbb{G}_m) \rightarrow H^1(R_{L/K}^{(1)}(\mathbb{G}_m)) \rightarrow H^1(R(\mathbb{G}_m)) \rightarrow H^1(\mathbb{G}_m).$$

The cohomology of $R(\mathbb{G}_m)$ and $\mathbb{G}_m$ is known: $H^0(C_{\acute{e}t}, R(\mathbb{G}_m)) = l^\times$, $H^0(C_{\acute{e}t}, \mathbb{G}_m) = k^\times$, $H^1(C_{\acute{e}t}, R(\mathbb{G}_m)) = \mathrm{Pic}(C_l)$, and $H^1(C_{\acute{e}t}, \mathbb{G}_m) = \mathrm{Pic}(C)$. Since the norm map from $l^\times$ to $k^\times$ is surjective, we have

$$H^1(C_{\acute{e}t}, R_{L/K}^{(1)}(\mathbb{G}_m)) \simeq \mathrm{Ker}[H^1(C_{\acute{e}t}, R(\mathbb{G}_m)) \rightarrow H^1(C_{\acute{e}t}, \mathbb{G}_m)].$$

Using the computation of the cohomology groups appearing in the kernel, we obtain

$$H^1(C_{\acute{e}t}, R_{L/K}^{(1)}(\mathbb{G}_m)) \simeq \mathrm{Ker} [N : \mathrm{Pic}(C_l) \rightarrow \mathrm{Pic}(C)].$$

From Theorems 6.1.1 and 6.1.2, we obtain

$$\mathrm{Cl}(\mathcal{T}) \simeq H^1(C_{\mathrm{cd}}, R_{L/K}^{(1)}(\mathbb{G}_m)) \simeq H^1(C_{\acute{e}t}, R_{L/K}^{(1)}(\mathbb{G}_m)).$$

This chain of isomorphisms proves the corollary. $\square$

6.3 Twisted Jacobians

Denote by J_C the Jacobian of the curve C . Since $\mathrm{Pic}(C_l) = J_C(l) \times \mathbb{Z}$, $\mathrm{Pic}(C) = J_C(k) \times \mathbb{Z}$, and the norm map between these groups acts on $\mathbb{Z}$ by multiplication by 2, we obtain

$$\mathrm{Ker}[N : \mathrm{Pic}(C_l) \rightarrow \mathrm{Pic}(C)] = \mathrm{Ker}[N : J_C(l) \rightarrow J_C(k)].$$

This kernel is the group of k -points of a certain abelian variety over the field k , called the $\widehat{T}$ -twisted Jacobian of the curve C (see [8]). Thus, by Corollary 6.2.2, we obtain that the group $\mathrm{Cl}(\mathcal{T})$ is the group of k -points of the twisted Jacobian of the curve C . Since these arguments do not depend on the base field k , we obtain that $\mathrm{Cl}(\mathcal{T}_l)$ is the group of l -points of the twisted Jacobian of the curve C .

A natural question arises: is it always true that for a constant torus $\mathcal{T}$ on a curve C , the group $\mathrm{Cl}(\mathcal{T})$ is the group of k -points of the $\widehat{T}$ -twisted Jacobian of the curve C ? We proved this above for norm tori; it is also true for tori obtained by Weil restriction of tori $\mathbb{G}_m^n$.

6.4 Computations

Consider the simplest nontrivial example, where the curve C is an elliptic curve over $\mathbb{F}_p$. Denote it by E , and let the torus $\mathcal{T}$ be the constant norm torus corresponding to the quadratic extension $\mathbb{F}_{p^2}/\mathbb{F}_p$. By the arguments in Subsection 6.3, the class groups of the torus T can be computed using points on a twisted elliptic curve ([8, Example 1.5(ii)]). Computations in PARI/GP [18] show that in this case the ℓ -primary part of the class group either does not grow, or grows with the asymptotic order of the class group at the n -th level equal to $\ell^{2n} + O(1)$, depending on whether the following polynomial is invertible in the ring $\mathbb{Z}_\ell[[T]]$:

$$1 + a_p(E)t + pt^2,$$

where $t = 1 + T$, and $a_p(E)$ is the trace of the Frobenius endomorphism acting on the Tate module of the elliptic curve E . This polynomial, viewed as a polynomial in the variable t , is the characteristic polynomial of Frobenius acting on the twisted Tate module.

In this case, on the basis of experimental data, one can formulate and prove the following theorem.

6.4.1 Theorem. *Let E be an elliptic curve over the prime field $k = \mathbb{F}_p$, let ℓ be an odd prime such that $p \equiv 1 \pmod{\ell}$, and let $\mathcal{T}$ be the constant norm torus corresponding to the quadratic extension $\mathbb{F}_{p^2}/\mathbb{F}_p$. Denote by k_n the field obtained from k by adjoining the ℓ^n -th roots of unity, and denote by E_n and $\mathcal{T}_n$ the curve obtained by extension of scalars of E to k_n , and the torus obtained by base change of $\mathcal{T}$ to E_n , respectively.*

Then, if ℓ does not divide the order of the group $\text{Cl}(\mathcal{T})$, then ℓ does not divide the orders of the groups $\text{Cl}(\mathcal{T}_n)$ for any n . If ℓ divides the order of the group $\text{Cl}(\mathcal{T})$, then ℓ divides the orders of the groups $\text{Cl}(\mathcal{T}_n)$ for every n . Moreover, in the second case the following refinement holds. Let m be the largest among the integers n such that $k_n = k$. For an arbitrary abelian group A , denote its ℓ -primary part by $A^{(\ell)}$. For $n \geq m$, the following divisibility holds:

$$\ell^2 \cdot \# \text{Cl}^{(\ell)}(\mathcal{T}_n) \mid \# \text{Cl}^{(\ell)}(\mathcal{T}_{n+1}).$$

Proof. We use the isomorphism between the group $\text{Cl}(\mathcal{T}_n)$ and the group of k_n -points on the twisted elliptic curve, which we denote by E' . By the Grothendieck–Lefschetz trace formula (see [10]), the k -points of E' can be described as follows:

$$\#E'(k) = p - a_p(E') + 1 = (1 - \alpha)(1 - \bar{\alpha}),$$

where $a_p(E') = \text{Tr}(\text{Frob}_p \mid T_\ell(E'))$ is the trace of the operator induced by the Frobenius morphism on the Tate module of the elliptic curve E' , α and $\bar{\alpha}$ are its eigenvalues, and the bar denotes complex conjugation. The k_n -points of E' are described similarly. To simplify the formulas, introduce the following notation:

$$q_n = p^{\ell^{(n-m)}}.$$

Then the following identity holds:

$$\#E'(k_n) = q_n - a_{q_n}(E') + 1 = (1 - \alpha_{q_n})(1 - \bar{\alpha}_{q_n}),$$

where $a_{q_n}(E') = \text{Tr}(\text{Frob}_{q_n} | T_\ell(E'))$ is the trace of the operator induced by the Frobenius morphism of degree q_n on the Tate module of the elliptic curve E' , and α_{q_n} and $\bar{\alpha}_{q_n}$ are its eigenvalues. From the following property of the Frobenius morphism,

$$\text{Frob}_{q_n} = \text{Frob}_p^{\ell^{(n-m)}},$$

we obtain the equality of eigenvalues

$$\alpha_{q_n} = \alpha^{\ell^{(n-m)}}.$$

To prove the first part of the theorem, observe that modulo ℓ we have the following congruence:

$$\#E'(k_n) = (1 - \alpha^{\ell^{(n-m)}})(1 - \bar{\alpha}^{\ell^{(n-m)}}) \equiv ((1 - \alpha)(1 - \bar{\alpha}))^{\ell^{(n-m)}} \equiv \#E'(k) \pmod{\ell}.$$

Using the identification of the class group $\text{Cl}(\mathcal{T}_n)$ with the group of points $E'(k_n)$, we obtain

$$\# \text{Cl}(\mathcal{T}_n) \equiv \# \text{Cl}(\mathcal{T}) \pmod{\ell}.$$

This proves the first part of the theorem. We now prove the refinement. Again we use the formula for the number of points in terms of the Frobenius eigenvalues:

$$\#E'(k_{n+1}) = (1 - \alpha^{\ell^{(n+1)}})(1 - \bar{\alpha}^{\ell^{(n+1)}}).$$

Put

$$a = \alpha^{\ell^n}.$$

We transform the right-hand side of the expression for the number of points:

$$(1 - a^\ell)(1 - \bar{a}^\ell) = (1 - a)(1 - \bar{a})(1 + a + \cdots + a^{\ell-1})(1 + \bar{a} + \cdots + \bar{a}^{\ell-1}).$$

The expression $(1 - a)(1 - \bar{a})$ is equal to the order of $\text{Cl}(\mathcal{T}_n)$, so it remains to prove that the remaining factor is divisible by ℓ^2 . Put $S(a) = (1 + a + \cdots + a^{\ell-1})$ and $b = a - 1$. Then

$$S(a) = \frac{a^\ell - 1}{a - 1} = \frac{(1 + b)^\ell - 1}{b} = \ell + C_\ell^2 \cdot b + \cdots + \ell \cdot b^{\ell-2} + b^{\ell-1} = \ell \cdot p(b) + b^{\ell-1},$$

for some integral polynomial p . There is an analogous expression for $S(\bar{a})$. Consider their product:

$$S(a)S(\bar{a}) = (\ell \cdot p(b) + b^{\ell-1})(\ell \cdot p(\bar{b}) + \bar{b}^{\ell-1}) = \ell^2 \cdot p(b)p(\bar{b}) + (\ell \cdot p(b)\bar{b}^{\ell-1} + \ell \cdot p(\bar{b})b^{\ell-1}) + (b\bar{b})^{\ell-1}.$$

Since $b\bar{b} = \# \text{Cl}(\mathcal{T}_n)$, by the part already proved this number is divisible by ℓ ; hence, being raised to the power $\ell - 1$, it is divisible at least by ℓ^2 . It remains to prove that the expression

$$\ell \cdot p(b)\bar{b}^{\ell-1} + \ell \cdot p(\bar{b})b^{\ell-1}$$

is divisible by ℓ^2 . The expression

$$p(b)\bar{b}^{\ell-1} + p(\bar{b})b^{\ell-1}$$

is symmetric with respect to the interchange of b and $\bar{b}$. Therefore, by the fundamental theorem on symmetric polynomials, it can be expressed in terms of polynomials in $b + \bar{b}$

and $b\bar{b}$. As already observed, $b\bar{b}$ is divisible by ℓ . We show that $b + \bar{b}$ is also divisible by ℓ . Consider the following expression:

$$(1 - a)(1 - \bar{a}) = 1 - (a + \bar{a}) + a\bar{a}.$$

Its left-hand side is divisible by ℓ , since it is equal to the order of $\text{Cl}(\mathcal{T}_n)$. One of the properties of the Frobenius eigenvalues is the following:

$$a\bar{a} = p^{\ell(n-m)}.$$

Since by assumption $p \equiv 1 \pmod{\ell}$, the following chain of congruences gives the desired divisibility:

$$0 \equiv (1 - a)(1 - \bar{a}) \equiv 2 - a - \bar{a} = -(b + \bar{b}) \pmod{\ell}.$$

□

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